

Accenture Computer Networking — Tier 3

Comprehensive Theory Guide, Comparative Reference, & 30 Solved MCQs with In-Depth Explanations

Part 1: Core Networking Concepts & Theory

1. Gateway ★★★★★

Definition

A gateway is a device or software function that connects different networks and can translate between different protocols or architectures. In many basic LAN questions, the term **default gateway** refers to the router/interface used to reach destinations outside the local subnet.

Traffic Flow Example

PC → Switch → Default Gateway → Router → Internet

If your computer wants to communicate outside its local network, it normally sends the packet toward the default gateway.

 **MCQ keyword:** Outside local network → Default Gateway

2. Repeater ★★★★★

Definition

A repeater receives a weakened signal, regenerates it, and retransmits it. It is associated with:

- **Physical Layer — Layer 1**

Example Flow

Computer → Weak signal → Repeater → Regenerated signal → Computer

Why use it?

To extend the physical transmission distance.

Important distinction — Repeater does NOT:

- Route packets
- Use IP addresses for routing
- Make intelligent MAC-based forwarding decisions

 **Remember:** Repeater → Regenerates signal → Layer 1

3. Bridge ★★★★★

Definition

A bridge connects network segments and forwards frames based on MAC addresses. It operates primarily at:

- **Layer 2 — Data Link**

Think of a bridge as an older/simple form of Layer-2 interconnection.

Example Topology

LAN A (PCs) → Bridge → LAN B (PCs)

Bridge vs Switch

Attribute	Bridge	Switch
OSI Layer	Layer 2 (Data Link)	Layer 2 (Data Link)
Function	Connects LAN segments	Connects many devices/segments
Technology	Older / software-based	Modern / hardware-based (ASIC)
Ports	Fewer ports traditionally	Many ports (high density)

 **Remember: Bridge → MAC → Layer 2**

4. Modem ★★★★★

Modem = Modulator-Demodulator

A modem converts signals between forms suitable for transmission over a communication medium and the digital data used by networking equipment.

Historically / Signal Conversion:

Digital → Modulation → Transmission medium → Demodulation → Digital

Common example

Your ISP provides a modem that connects your home network to the ISP's access network.

Important Note

Don't confuse: **Modem ≠ Router**

A modem provides the interface to the ISP's access technology; a router directs traffic between networks. Modern home devices often combine: **Modem + Router + Wi-Fi access point** into one device.

5. Proxy Server ★★★★★

Definition

A proxy server acts as an intermediary between a client and another server. Instead of Client → Internet Server, you have:

Client → Proxy → Internet Server

The proxy can:

- Filter requests
- Cache content
- Enforce access policies
- Hide the client's direct network connection from the destination in some architectures

Example: Corporate Filtering

Employee → Proxy → Check policy → Allow / Block → Internet

 **Keyword: Intermediary → Proxy**

6. Firewall ★★★★★

Definition

A firewall is a security control that monitors and controls network traffic according to defined security rules.

Internet → Firewall → Company Network

Traffic Filtering Criteria:

- Source IP & Destination IP
- Port number & Protocol (TCP/UDP)
- Application, depending on firewall type

Example Rule: Block TCP port 23 (prevents unencrypted Telnet traffic from passing through the firewall).

Firewall vs Router

- **Router:** Determines where packets should go.
- **Firewall:** Determines whether traffic should be allowed according to security policy.
- *Note:* A single hardware device can perform both routing and firewall functions.

7. VPN ★★★★★

Definition

VPN = Virtual Private Network. A VPN creates a protected communication path over an untrusted/public network.

Employee → Internet → Encrypted VPN connection → Company Network

Why VPN?

- Secure remote access
- Protect traffic over untrusted networks
- Connect remote networks securely

VPN vs Proxy — Key Distinction

- **Proxy:** Usually acts as an intermediary for particular application traffic (Client → Proxy → Server).
- **VPN:** Typically creates a protected tunnel for all network traffic between endpoints (Device → Encrypted VPN tunnel → Company network).

🧠 Application intermediary → Proxy | Secure network tunnel → VPN

8. Routing Basics ★★★★★

What is routing?

Routing is the process of determining a path for packets to travel from source to destination across networks.

Network A → Router 1 → Router 2 → Network B

Routers use **routing tables** to make forwarding decisions.

Routing Table Structure

A routing table contains information such as: *Destination Network, Next Hop, Interface, Metric.*

Destination Network	Next Hop	Notes
192.168.2.0/24	10.0.0.2	Specific subnet route
0.0.0.0/0	10.0.0.1	Default route

Default Route (0.0.0.0/0)

A default route is commonly represented as $0.0.0.0/0$. It means: *Use this route when no more specific route matches.*

🧠 Keyword: No specific route → Default route (0.0.0.0/0)

9. Static vs Dynamic Routing ★★★★★

Static Routing

Routes are manually configured by an administrator (Administrator → Manual route → Router).

- **Advantages:** Simple for small networks, predictable, no routing-protocol overhead.
- **Disadvantages:** Manual configuration, difficult to maintain in large networks, doesn't automatically adapt to topology changes.

Dynamic Routing

Routers use routing protocols to learn routes automatically (e.g., **OSPF, RIP, BGP**).

- **Advantages:** Automatically learns routes, adapts to topology changes, scales well for larger networks.
- **Disadvantages:** More complex, uses CPU/memory/bandwidth for protocol exchange.

Feature	Static Routing	Dynamic Routing
Configuration	Manually configured	Learned through routing protocols
Complexity	Simple	More complex
Network Size	Good for small/stable networks	Good for larger/changing networks
Adaptability	Doesn't automatically adapt	Can adapt dynamically

10. Anycast ★★★★★

Addressing comparison review:

- **Unicast:** One → One specific destination
- **Broadcast:** One → Everyone in the broadcast domain
- **Multicast:** One → Selected group of subscribers

Anycast Concept

One sender sends traffic to **one destination selected from a group of possible destinations**, typically based on routing/metric (e.g. topologically nearest instance).



Common use: Widely used in distributed services such as DNS root servers and CDNs (Content Delivery Networks).

 **Memory:** Unicast → One | Broadcast → Everyone | Multicast → Group | Anycast → One of several

11. Collision Domain ★★★ & 12. Broadcast Domain ★★★★★

Collision Domain

A network segment where simultaneous transmissions can potentially interfere/collide in traditional shared-medium Ethernet.

- **Hub:** Creates **one shared collision domain** for all connected ports.
- **Switch:** Each switch port is typically its own **separate collision domain** in modern full-duplex Ethernet.

Broadcast Domain

The set of devices that receive a Layer-2 broadcast frame.

- **Switch:** A normal Layer-2 switch forwards broadcasts within the same VLAN (does not separate broadcast domains).
- **Router:** A router normally **separates broadcast domains** across its interfaces.

🧠 **Key Summary:** Hub → 1 Collision Domain | Switch → Separates Collision Domains | Router → Separates Broadcast Domains

13. Network Commands Cheat Sheet ★★★

Command	Purpose / Description	Example / Syntax
ping	Tests basic IP reachability using ICMP Echo messages	ping 8.8.8.8
ipconfig	Windows command to view IP, subnet mask, default gateway	ipconfig /all
ifconfig / ip addr	Unix/Linux command for interface configuration/status	ip addr
tracert / traceroute	Shows hop-by-hop path taken by packets toward destination	tracert google.com (Win) traceroute google.com (Linux)
nslookup	Queries DNS servers for name-to-IP resolution	nslookup google.com

14. Bandwidth ★★★ | 15. Throughput ★★★ | 16. Latency ★★★★★

- **Bandwidth:** Maximum data-carrying capacity of a link (e.g., 100 Mbps, 1 Gbps).
- **Throughput:** Actual rate of successful data transfer achieved over a network (e.g., Bandwidth = 100 Mbps, Throughput = 75 Mbps due to congestion, packet loss, protocol overhead).
- **Latency:** Time taken for data to travel from source to destination, measured in milliseconds (ms). Essential for online gaming and real-time voice.

The Highway Analogy

Bandwidth → Number of highway lanes (maximum capacity)
Throughput → Actual number of cars passing per hour (achieved rate)
Latency → Time taken for a single car to reach its destination (delay)

17. Packet Switching ★★★★★ vs 18. Circuit Switching ★★★★★

Attribute	Packet Switching	Circuit Switching
Data Handling	Data divided into discrete packets	Continuous stream of data
Path Setup	No dedicated path required (packets can take different paths)	Dedicated communication path established for entire session
Efficiency	Efficient for bursty traffic; shared resources	Dedicated resources; idle capacity is wasted
Primary Use	The Internet	Traditional telephone networks (PSTN)

 Accenture keyword: Internet → Packet switching

Part 2: Practice MCQs with Inline Explanations

Q1. Gateway

A PC wants to communicate with a server located outside its local subnet. Which device/interface does it normally send the packet toward first?

- A. DNS server
- B. Default gateway
- C. DHCP server
- D. Repeater

Answer: B. Default gateway ✓

When a destination is outside the local subnet, the host normally forwards the packet to its default gateway.

- A. DNS server ✗ — Resolves domain names to IP addresses.
- B. Default gateway ✓ — Provides the next-hop path to other networks.
- C. DHCP server ✗ — Provides dynamic IP configuration.
- D. Repeater ✗ — Regenerates physical signals.

 Outside local network → Default Gateway

Q2. Repeater

A network administrator installs a device that receives a weak signal, regenerates it, and retransmits it. What device is this?

- A. Router
- B. Switch
- C. Repeater
- D. Gateway

Answer: C. Repeater ✓

A repeater receives a weakened signal, regenerates it, and retransmits it.

- A. Router ✗ — Routes packets between networks.
- B. Switch ✗ — Forwards frames based primarily on MAC addresses.
- C. Repeater ✓ — Regenerates/repeats signals.
- D. Gateway ✗ — Connects networks/systems and may perform protocol translation.

🧠 Weak signal → Regenerate → Repeater

Q3. Repeater Layer

A repeater primarily operates at which OSI layer?

- A. Physical
- B. Data Link
- C. Network
- D. Transport

Answer: A. Physical ✓

A repeater works at: Layer 1 — Physical.

- A. Physical ✓ — Deals with physical signals and raw bits.
- B. Data Link ✗ — Frames and MAC addresses.
- C. Network ✗ — IP packets and routing.
- D. Transport ✗ — TCP/UDP segments and port numbers.

Q4. Bridge

Which device primarily forwards frames based on MAC addresses and connects network segments?

- A. Router
- B. Bridge
- C. Repeater
- D. Modem

Answer: B. Bridge ✓

A bridge connects LAN segments and forwards frames based on MAC addresses.

- A. Router ✗ — Primarily routes using IP.
- B. Bridge ✓ — Layer-2/MAC-based forwarding.
- C. Repeater ✗ — Regenerates signals at Layer 1.
- D. Modem ✗ — Converts/modulates signals for communication over an access medium.

🧠 Bridge → MAC → Layer 2

Q5. Bridge vs Router

Which statement is correct?

- A. Bridge primarily works with MAC addresses, while router primarily uses IP addresses
- B. Bridge routes packets between IP networks, while router only regenerates signals
- C. Bridge operates at Layer 4, while router operates at Layer 1
- D. Bridge and router perform exactly the same function

Answer: A. Bridge primarily works with MAC addresses, while router primarily uses IP addresses ✓

- A. ✓ *Correct — Bridge operates at Layer 2 (MAC); Router operates at Layer 3 (IP).*
- B. ✗ *Roles are reversed.*
- C. ✗ *Bridge is Layer 2; router is primarily Layer 3.*
- D. ✗ *They perform distinct networking functions.*

Remember: Bridge → MAC → Layer 2 | Router → IP → Layer 3

Q6. Modem

What does the term modem originate from?

- A. Monitor-Demonstrator
- B. Modulator-Demodulator
- C. Module-Domain
- D. Modern-Digital

Answer: B. Modulator-Demodulator ✓

The word modem comes from: MODulator + DEModulator.

A. Monitor-Demonstrator ✗ | B. Modulator-Demodulator ✓ | C. Module-Domain ✗ | D. Modern-Digital ✗

Q7. Proxy

An organization routes employees' web requests through an intermediary that can inspect requests, enforce access policies, and cache content. What is this intermediary?

- A. Proxy server
- B. Repeater
- C. Modem
- D. DHCP server

Answer: A. Proxy server ✓

A proxy acts as an intermediary: Client → Proxy → Internet Server.

- A. Proxy server ✓ *Intermediary that can filter/cache/forward requests.*
- B. Repeater ✗ *Regenerates signals.*
- C. Modem ✗ *Provides communication-medium/ISP access functionality.*
- D. DHCP server ✗ *Assigns network configuration dynamically.*

🧠 **Intermediary → Proxy**

Q8. VPN

An employee working from home needs secure access to an internal company application over the Internet. Which technology is most appropriate?

- A. DNS
- B. VPN
- C. DHCP
- D. ARP

Answer: B. VPN ✓

An employee needs secure remote access to an internal company resource over the Internet.

- A. DNS ✗ — Name resolution.
- B. VPN ✓ — Provides a protected connection/tunnel for remote access.
- C. DHCP ✗ — IP configuration.
- D. ARP ✗ — IPv4-to-MAC address resolution on the local network.

🧠 Remote + secure company network → VPN

Q9. Proxy vs VPN

Which statement is most accurate?

- A. A proxy can act as an intermediary for application traffic, while a VPN typically provides a protected network connection/tunnel
- B. A VPN only translates domain names, while a proxy assigns IP addresses
- C. Proxy and VPN always perform exactly the same function
- D. A proxy is used only for physical-layer communication

Answer: A. A proxy can act as an intermediary for application traffic, while a VPN typically provides a protected network connection/tunnel ✓

- A. ✓ Correct distinction.
- B. ✗ — DNS resolves names; DHCP provides configuration.
- C. ✗ — They operate differently (application vs network tunnel).
- D. ✗ — Proxy isn't a Physical-layer technology.

Easy distinction: Proxy → Intermediary | VPN → Protected network connection/tunnel

Q10. Firewall

Which security control primarily allows or blocks network traffic based on configured rules?

- A. Firewall
- B. DNS
- C. DHCP
- D. Repeater

Answer: A. Firewall ✓

A firewall controls traffic according to security rules.

- A. Firewall ✓ — Allows/blocks traffic based on policy.
- B. DNS ✗ — Resolves domain names to IP addresses.
- C. DHCP ✗ — Assigns network IP configuration.
- D. Repeater ✗ — Regenerates physical signals.

Q11. Firewall vs Router

Which statement is correct?

- A. Router primarily decides where packets should be forwarded; firewall primarily controls whether traffic is permitted according to security policy
- B. Router encrypts all data, while firewall assigns IP addresses
- C. Firewall performs only signal regeneration
- D. Router and firewall are exactly the same device/function

Answer: A. Router primarily decides where packets should be forwarded; firewall primarily controls whether traffic is permitted according to security policy ✓

- A. ✓ *Correct.*
- B. ✗ — Router doesn't primarily encrypt traffic; firewall doesn't assign IP addresses.
- C. ✗ — Signal regeneration is a repeater's function.
- D. ✗ — Different functions (though often co-located in unified security devices).

Q12. Routing

What is the main purpose of routing?

- A. Convert domain names into IP addresses
- B. Determine paths for packets between networks
- C. Encrypt application data
- D. Assign MAC addresses

Answer: B. Determine paths for packets between networks ✓

- A. ✗ — DNS function.
- B. ✓ *Routing — Path determination across networks.*
- C. ✗ — Encryption/Security protocols.
- D. ✗ — MAC addressing is hardware-bound at Layer 2.

 **Routing = Finding a path**

Q13. Routing Table

Which information would normally be found in a router's routing table?

- A. Destination network and next hop
- B. User passwords
- C. Email messages
- D. Browser history

Answer: A. Destination network and next hop ✓

A routing table can contain information such as: Destination network, Next hop, Interface, Metric.

- A. ✓ *Correct.*
- B. ✗ — Passwords aren't routing information.
- C. ✗ — Email isn't routing-table data.
- D. ✗ — Browser history isn't routing information.

Q14. Default Route

In IPv4, which prefix commonly represents the default route?

- A. 255.255.255.255/32
- B. 127.0.0.0/8
- C. 0.0.0.0/0
- D. 192.168.0.0/16

Answer: C. 0.0.0.0/0 ✓

0.0.0.0/0 represents the IPv4 default route. If no more-specific route matches, use this route.

A. 255.255.255.255/32 ✗ — Limited broadcast address.

B. 127.0.0.0/8 ✗ — IPv4 loopback range.

C. 0.0.0.0/0 ✓ — Default route.

D. 192.168.0.0/16 ✗ — Private IPv4 address block.

🧠 **Default route** → 0.0.0.0/0

Q15. Static Routing

An administrator manually configures a route on a router. What type of routing is this?

- A. Dynamic routing
- B. Static routing
- C. Multicast routing only
- D. Broadcast routing

Answer: B. Static routing ✓

The administrator manually configures the route.

A. Dynamic routing ✗ — Routes are learned through routing protocols.

B. Static routing ✓ — Manually configured.

C. Multicast routing only ✗ | D. Broadcast routing ✗

Q16. Dynamic Routing

Routers automatically learn and update routes using routing protocols. What type of routing is this?

- A. Static routing
- B. Dynamic routing
- C. Manual routing
- D. Physical routing

Answer: B. Dynamic routing ✓

Routers automatically learn/update routes using routing protocols.

A. Static routing ✗ — Manual.

B. Dynamic routing ✓

C. Manual routing ✗ — Describes static routing.

D. Physical routing ✗ — Not a standard routing category.

🧠 **Automatic route learning** → Dynamic routing

Q17. Routing Protocol

Which of the following is a dynamic routing protocol?

- A. HTTP
- B. OSPF
- C. DNS
- D. FTP

Answer: B. OSPF ✓

OSPF = Open Shortest Path First. It is an interior dynamic routing protocol.

A. HTTP ✗ — Web protocol (Layer 7).

B. OSPF ✓ — Dynamic routing protocol (Layer 3).

C. DNS ✗ — Name resolution protocol.

D. FTP ✗ — File transfer protocol.

Other important routing protocols: RIP, EIGRP, BGP.

Q18. Anycast

A service is deployed on several servers. A client sends traffic to an address that can be routed to one appropriate server among several possible instances. What addressing/communication concept best describes this?

- A. Broadcast
- B. Multicast
- C. Anycast
- D. Flooding

Answer: C. Anycast ✓

A service is available from multiple servers, and routing selects one appropriate destination.

A. Broadcast ✗ — One → all.

B. Multicast ✗ — One → selected group.

C. Anycast ✓ — One → one of multiple possible destinations.

D. Flooding ✗ — Sending traffic broadly through a network.

🧠 **Anycast = One of many**

Q19. Unicast

A client sends a request to exactly one web server. What type of communication is this?

- A. Broadcast
- B. Multicast
- C. Unicast
- D. Anycast

Answer: C. Unicast ✓

One client communicates with exactly one destination server.

A. Broadcast ✗ — One → all in the broadcast domain.

B. Multicast ✗ — One → group.

C. Unicast ✓ — One → one.

D. Anycast ✗ — One → one selected from multiple possible destinations.

Q20. Broadcast

One device sends a Layer-2 broadcast frame on a LAN. Who is intended to receive it?

- A. Only the sender
- B. One specific device
- C. All devices in the relevant broadcast domain
- D. Only the router on the Internet

Answer: C. All devices in the relevant broadcast domain ✓

A. Only sender ✗ | B. One specific device ✗ — That's unicast.

C. All devices in the broadcast domain ✓

D. Only Internet router ✗

 **Broadcast = One → All within broadcast domain**

Q21. Multicast

A streaming service sends traffic to a specific group of subscribed receivers. Which communication type is appropriate?

- A. Unicast
- B. Broadcast
- C. Multicast
- D. Anycast

Answer: C. Multicast ✓

A stream is sent to a specific group of subscribed receivers.

A. Unicast ✗ — One receiver.

B. Broadcast ✗ — All devices in the broadcast domain.

C. Multicast ✓ — Selected group.

D. Anycast ✗ — One destination selected from multiple possible destinations.

Q22. Collision Domain

Which device traditionally creates a single shared collision domain for all connected devices?

- A. Hub
- B. Router
- C. Switch
- D. Firewall

Answer: A. Hub ✓

A traditional Ethernet hub creates one shared collision domain.

A. Hub ✓

B. Router ✗ — Separates broadcast domains and collision domains per interface.

C. Switch ✗ — Each switch port is typically a separate collision domain.

D. Firewall ✗ — Security filter.

 **Hub → One shared collision domain**

Q23. Switch and Collision Domains

In a modern Ethernet network using a switch, each switch port is typically:

- A. A separate collision domain
- B. A separate Internet
- C. A separate broadcast domain regardless of VLAN configuration
- D. A separate DNS domain

Answer: A. A separate collision domain ✓

In modern switched Ethernet, each switch port is typically a separate collision domain.

A. *Separate collision domain* ✓

B. *Separate Internet* ✗

C. *Separate broadcast domain regardless of VLAN* ✗ — Normal switch does not separate broadcast domains per port (requires VLANs).

D. *Separate DNS domain* ✗

Important: Switch separates collision domains, not automatically broadcast domains.

Q24. Broadcast Domain

Which device normally separates Layer-2 broadcast domains?

- A. Hub
- B. Repeater
- C. Router
- D. Ethernet cable

Answer: C. Router ✓

A router normally separates Layer-2 broadcast domains.

A. *Hub* ✗ — Extends the same broadcast domain.

B. *Repeater* ✗ — Regenerates signals.

C. *Router* ✓ — Does not normally forward Layer-2 broadcasts between interfaces.

D. *Ethernet cable* ✗

 **Router → Broadcast domain boundary**

Q25. Ping

A network engineer wants to test basic IP-level reachability to a remote host. Which command is most appropriate?

- A. ping
- B. nslookup
- C. ipconfig
- D. tracert

Answer: A. ping ✓

ping commonly uses ICMP Echo messages to test IP-level reachability.

A. *ping* ✓

B. *nslookup* ✗ — DNS lookup.

C. *ipconfig* ✗ — Displays local network configuration on Windows.

D. *tracert* ✗ — Shows the hop-by-hop path.

Q26. DNS Troubleshooting

A user knows a website's domain name but wants to check what IP address DNS resolves it to. Which command is most appropriate?

- A. ping
- B. nslookup
- C. ipconfig
- D. tracert

Answer: B. nslookup ✓

If you want to check what IP address a domain resolves to: `nslookup google.com`.

A. **ping** ✗ — Tests reachability (may show IP, but is not the primary DNS query tool).

B. **nslookup** ✓ — Primary DNS query tool.

C. **ipconfig** ✗ — Local network configuration.

D. **tracert** ✗ — Traces network path.

🧠 **DNS lookup → nslookup**

Q27. Route Troubleshooting

A network engineer wants to see the sequence of hops packets take toward a destination. Which command is appropriate?

- A. nslookup
- B. ipconfig
- C. tracert / traceroute
- D. ping only

Answer: C. tracert / traceroute ✓

These commands show the hops toward a destination.

A. **nslookup** ✗ — DNS queries.

B. **ipconfig** ✗ — Local configuration.

C. **tracert / traceroute** ✓ — Shows network path/hops.

D. **ping only** ✗ — Tests reachability without detailed per-hop analysis.

Q28. Bandwidth vs Throughput

A network link has a theoretical capacity of 1 Gbps, but actual data transfer is only 700 Mbps. Which statement is correct?

- A. Bandwidth is 700 Mbps and throughput is 1 Gbps
- B. Bandwidth is 1 Gbps and throughput is 700 Mbps
- C. Both are necessarily 700 Mbps
- D. Neither can be measured in bits per second

Answer: B. Bandwidth is 1 Gbps and throughput is 700 Mbps ✓

Maximum capacity = 1 Gbps (Bandwidth); Actual transfer achieved = 700 Mbps (Throughput).

A. ✗ *Reversed.*

B. ✓ *Correct.*

C. ✗ *They measure different concepts.*

D. ✗ *Both are measured in bps (bits per second).*

🧠 **Bandwidth = capacity | Throughput = actual performance**

Q29. Latency

An online game becomes difficult to play because responses take a long time to arrive even though the connection has high bandwidth. Which metric is most directly associated with this delay?

- A. MAC address
- B. Latency
- C. Subnet mask
- D. Port number

Answer: B. Latency ✓

Latency represents the delay in communication.

A. MAC address ✗ — Layer-2 address.

B. Latency ✓ — Communication delay.

C. Subnet mask ✗ — Defines network/host portions.

D. Port number ✗ — Identifies a service/application endpoint.

🧠 High delay → High latency

Q30. 🔥 Mixed Scenario

A company has the following network design:

- A repeater extends a physical signal.
- A switch forwards Ethernet frames using MAC addresses.
- A router forwards packets between networks.
- A firewall controls traffic according to security rules.
- A VPN provides secure remote connectivity.
- OSPF dynamically learns routes.

Which statement is correct?

- A. Repeater → Layer 1, Switch → Layer 2, Router → Layer 3, OSPF → Dynamic routing
- B. Repeater → Layer 3, Switch → Layer 4, Router → Layer 1, OSPF → Application protocol
- C. Repeater → Layer 2, Switch → Layer 1, Router → Layer 4, OSPF → Email protocol
- D. All devices operate at the same OSI layer

Answer: A. Repeater → Layer 1, Switch → Layer 2, Router → Layer 3, OSPF → Dynamic routing ✓

Repeater → Layer 1 (Signal regeneration)

Switch → Layer 2 (MAC / frames)

Router → Layer 3 (IP / routing)

OSPF → Dynamic routing protocol

A. ✓ Correct mapping.

B. ✗ — Layers are incorrect and OSPF isn't an Application protocol.

C. ✗ — Layers and protocol functions are incorrect.

D. ✗ — These devices/protocols operate at different layers/functions.